

QJT2

Calculus I

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A physics teacher asks her class of 28 to graph $f(x) = \sqrt{(x - 3) / (x - 7)}$ on calculators set to a window of -10 to 10 . Before graphing, students must state the domain. What is it?

- A. All real numbers except 7
- B. $x \geq 3$ and $x \neq 7$
- C. $x \leq 3$
- D. $x > 3$ and $x \neq 7$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A toll authority models its price with $f(x) = x^2 + 1$ for $x < 2$ and $f(x) = 3x - 1$ for $x > 2$, where x is miles. A clerk entered $f(2) = 9$ after a system update in March. What is the limit of $f(x)$ as x approaches 2?

- A. The limit does not exist because $f(2)$ differs from nearby values.
- B. 5
- C. 9
- D. The limit does not exist because a piecewise function always breaks at the switch point.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A shipping company's cost model is $f(x) = |x - 4| + 2$, where x is days late. The model was built by an analyst team of six in about two weeks. At $x = 4$, which statement about f is true?

- A. f is neither continuous nor differentiable at $x = 4$.
- B. $f'(4) = 0$ because $x = 4$ is the minimum.
- C. f is continuous but not differentiable at $x = 4$.
- D. f is differentiable at $x = 4$ because it is continuous there.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A function f is known to be differentiable at $x = a$. Which statement must also be true?

- A. The limit of $f(x)$ as $x \rightarrow a$ may differ from $f(a)$.
- B. $f(a)$ is a local maximum or minimum.
- C. f is continuous at $x = a$.
- D. f is differentiable at every x near a .

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A calculus coach preparing a team of 12 for a spring contest writes the limit of $[(2 + h)^3 - 8] / h$ as $h \rightarrow 0$ on the board. Last year the team placed third. What is the value of the limit?

- A. 12
- B. 0, because the numerator goes to 0.
- C. 6
- D. 8

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

The limit of $[\sqrt{9 + h} - 3] / h$ as $h \rightarrow 0$ is the derivative of which function at which point, and what is its value?

- A. $f(x) = \sqrt{x}$ at $x = 9$; value $1/6$
- B. $f(x) = \sqrt{x}$ at $x = 9$; value $1/3$
- C. $f(x) = \sqrt{x}$ at $x = 3$; value $1/6$
- D. $f(x) = \sqrt{x}$ at $x = 9$; value 6

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A publisher models printing cost with $g(x) = 4x^3 - 5x^2 + 7x - 9$, fit to data from a three-year pilot that used 11 print shops. What is the instantaneous rate of change of g at $x = 2$?

- A. 75
- B. 35
- C. 26
- D. 17

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A fish hatchery that employs 14 staff models its trout count as $P(t) = 500e^{(0.04t)}$, where t is days. Use $e^{0.4} \approx 1.49$. The hatchery's tanks hold 20,000 liters each. How fast is the population growing at $t = 10$ days?

- A. About 298 trout per day
- B. 20 trout per day
- C. About 745 trout per day
- D. About 29.8 trout per day

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

An engineer with a new graphing tablet wants the tangent line to $f(x) = x^2 - 3x + 1$ at $x = 4$. Her supervisor has asked for the result by Friday. Which equation is the tangent line?

- A. $y = 2x - 3$
- B. $y = 5x$
- C. $y = 5x - 15$
- D. $y = 5x + 25$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

The tangent line to a function f at $x = 2$ is $y = 5x - 3$. What are $f(2)$ and $f'(2)$?

- A. $f(2) = 7$ and $f'(2) = 5$
- B. $f(2) = 7$ and $f'(2) = 0$
- C. $f(2) = 5$ and $f'(2) = 7$
- D. $f(2) = -3$ and $f'(2) = 5$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A market analyst with 15 years of experience models average cost as $C(x) = (6x^2 - x) / (3x^2 + 5x + 2)$, where x is thousands of units. The client is a furniture maker. What value does $C(x)$ approach as $x \rightarrow \infty$?

- A. It grows without bound.
- B. $-1/5$
- C. 0
- D. 2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

The function $f(x) = (kx^3 + 1) / (4x^3 - x)$ has a horizontal asymptote $y = 3$. What is k ?

- A.** 3
- B.** No k works, because equal degrees give an asymptote of 0.
- C.** $3/4$
- D.** 12

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A physics teacher asks ~~her class of 28~~ to graph $f(x) = \sqrt{x - 3} / (x - 7)$ on calculators ~~set to a window of -10 to 10~~ . Before graphing, students must state the domain. What is it?

Target word: domain **Decides:** Need $x - 3 \geq 0$ and $x - 7 \neq 0$.

A	Dropped element	Drops the square-root restriction.
B	Correct	The radicand must be nonnegative and the denominator nonzero: $[3, 7) \cup (7, \infty)$.
C	Inverted operation	Flips the inequality for the radicand.
D	Adjacent category	Excludes 3, but $\sqrt{0} = 0$ is defined, so 3 belongs in the domain.

2. Answer B Concept 2

A toll authority models its price with $f(x) = x^2 + 1$ for $x < 2$ and $f(x) = 3x - 1$ for $x > 2$, where x is miles. A clerk entered $f(2) = 9$ ~~after a system update in March~~. What is the limit of $f(x)$ as x approaches 2?

Target word: two-sided limit at 2 **Decides:** Left limit $2^2 + 1 = 5$; right limit $3(2) - 1 = 5$.

A	Adjacent category	That makes f discontinuous at 2, but the limit still exists.
B	Correct	Both one-sided limits equal 5; the value at 2 does not matter.
C	Confusion set	That is the function value $f(2)$, not the limit.
D	Absolute language	Piecewise pieces can meet; here both sides approach 5.

3. Answer C Concept 3

A shipping company's cost model is $f(x) = |x - 4| + 2$, where x is days late. ~~The model was built by an analyst team of six in about two weeks.~~ At $x = 4$, which statement about f is true?

Target word: continuity vs. differentiability at a corner **Decides:** The graph has a V-shaped corner at $x = 4$: slopes -1 on the left and $+1$ on the right.

A	Adjacent category	The two pieces meet at $(4, 2)$, so f is continuous.
B	Confusion set	Borrows the smooth-minimum rule; at a corner the derivative does not exist.
C	Correct	No break in the graph, but the left and right slopes disagree.
D	Inverted operation	Reverses the implication; continuity does not guarantee a derivative.

4. Answer C Concept 3 · transfer

A function f is known to be differentiable at $x = a$. Which statement must also be true?

Target word: what differentiability guarantees **Decides:** Differentiable at a implies continuous at a .

A	Inverted operation	That describes a discontinuity, which differentiability rules out.
B	Confusion set	Extrema need $f'(a) = 0$ or undefined; differentiability alone says nothing about extrema.
C	Correct	Differentiability at a point implies continuity there.
D	Absolute language	Differentiability at one point says nothing about nearby points.

5. Answer A Concept 4

A calculus coach ~~preparing a team of 12 for a spring contest~~ writes the limit of $[(2 + h)^3 - 8] / h$ as $h \rightarrow 0$ on the board. ~~Last year the team placed third.~~ What is the value of the limit?

Target word: value of a derivative written as a limit **Decides:** The expression is $f'(2)$ for $f(x) = x^3$, and $f'(x) = 3x^2$.

A	Correct	$f'(2) = 3(2)^2 = 12$.
B	Dropped element	Drops the denominator, which also goes to 0; the form is 0/0, not 0.
C	Confusion set	Uses $3x$ instead of $3x^2$ for the derivative of x^3 .
D	Wrong stem number	That is $f(2) = 2^3$, not the slope.

6. Answer A Concept 4 • transfer

The limit of $[\sqrt{9 + h} - 3] / h$ as $h \rightarrow 0$ is the derivative of which function at which point, and what is its value?

Target word: identify f and a , then evaluate **Decides:** Pattern $[f(a + h) - f(a)] / h$ with $f(x) = \sqrt{x}$ and $a = 9$; $f'(x) = 1 / (2\sqrt{x})$.

A	Correct	$1 / (2\sqrt{9}) = 1/6$.
B	Dropped element	Drops the factor $1/2$ in the derivative of $\sqrt{x}$.
C	Wrong stem number	3 is $f(9)$, not the input a .
D	Inverted operation	Flips the fraction.

7. Answer B Concept 5

A publisher models printing cost with $g(x) = 4x^3 - 5x^2 + 7x - 9$, ~~fit to data from a three-year pilot that used 11 print shops.~~ What is the instantaneous rate of change of g at $x = 2$?

Target word: $g'(2)$ **Decides:** $g'(x) = 12x^2 - 10x + 7$; the constant term drops out.

A	Inverted operation	Sign error on the middle term: $48 + 20 + 7$.
B	Correct	$12(4) - 10(2) + 7 = 48 - 20 + 7 = 35$.
C	Dropped element	Keeps the -9 ; the derivative of a constant is 0.
D	Confusion set	That is $g(2) = 32 - 20 + 14 - 9$, the function value.

8. Answer D Concept 6

A fish hatchery ~~that employs 14 staff~~ models its trout count as $P(t) = 500e^{(0.04t)}$, where t is days. Use $e^{0.4} \approx 1.49$. ~~The hatchery's tanks hold 20,000 liters each.~~ How fast is the population growing at $t = 10$ days?

Target word: $P'(10)$ **Decides:** $P'(t) = 0.04 \times 500e^{(0.04t)} = 20e^{(0.04t)}$.

A	Wrong stem number	Uses 0.4 instead of the growth constant 0.04.
B	Dropped element	Drops $e^{(0.04t)}$; that is the rate at $t = 0$.
C	Confusion set	$500 \times 1.49 \approx 745$ is $P(10)$, the count, not the rate.
D	Correct	$20 \times e^{0.4} \approx 20 \times 1.49 = 29.8$.

9. Answer C Concept 7

An engineer ~~with a new graphing tablet~~ wants the tangent line to $f(x) = x^2 - 3x + 1$ at $x = 4$. ~~Her supervisor has asked for the result by Friday.~~ Which equation is the tangent line?

Target word: tangent line equation **Decides:** $f(4) = 5$ and $f'(x) = 2x - 3$ gives $f'(4) = 5$.

A	Confusion set	That is the derivative formula, not the tangent line.
B	Dropped element	Has the right slope but drops the point (4, 5).
C	Correct	$y = 5 + 5(x - 4) = 5x - 15$.
D	Inverted operation	Uses $(x + 4)$ instead of $(x - 4)$.

10. Answer A Concept 7 • transfer

The tangent line to a function f at $x = 2$ is $y = 5x - 3$. What are $f(2)$ and $f'(2)$?

Target word: read $f(a)$ and $f'(a)$ back from the tangent line **Decides:** Slope is $f'(2)$; the line's value at $x = 2$ is $f(2)$.

A	Correct	$5(2) - 3 = 7$, and the slope is 5.
B	Confusion set	A tangent line is not always horizontal.
C	Inverted operation	Swaps the function value and the slope.
D	Wrong stem number	-3 is the y-intercept, not the value at $x = 2$.

11. Answer D Concept 8

A market analyst ~~with 15 years of experience~~ models average cost as $C(x) = (6x^2 - x) / (3x^2 + 5x + 2)$, where x is thousands of units. ~~The client is a furniture maker.~~ What value does $C(x)$ approach as $x \rightarrow \infty$?

Target word: limit at infinity **Decides:** Equal degrees: ratio of leading coefficients $6/3$.

A	Confusion set	That happens only when the numerator's degree is larger.
B	Dropped element	Uses the x-terms $-x / 5x$ and drops the leading terms.
C	Adjacent category	That rule applies when the numerator's degree is smaller.
D	Correct	$6x^2 / 3x^2 = 2$.

12. Answer D Concept 8 • transfer

The function $f(x) = (kx^3 + 1) / (4x^3 - x)$ has a horizontal asymptote $y = 3$. What is k ?

Target word: solve for a coefficient from the asymptote **Decides:** Equal degrees, so $k / 4 = 3$.

A	Wrong stem number	That is the asymptote value, not k .
B	Adjacent category	A 0 asymptote belongs to a smaller numerator degree.
C	Inverted operation	Divides 3 by 4 instead of multiplying.
D	Correct	$k / 4 = 3$ gives $k = 12$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Domain of a function	The domain excludes inputs that make a denominator zero or an even root's radicand negative; both restrictions apply at once.	Applying only one restriction, or excluding an endpoint where the radicand equals zero.	A square root and a fraction in the same formula.
2	One-sided limits and limit existence	The limit at a exists exactly when the left and right limits exist and agree; the value $f(a)$ does not affect it.	Reporting $f(a)$ as the limit, or saying no limit exists because the formula changes.	A piecewise function with a separately defined value at the break point.
3	Differentiability implies continuity	A function differentiable at a is continuous at a , but continuity does not guarantee differentiability (corners, cusps, vertical tangents).	Reversing the implication: "continuous, so differentiable."	An absolute value or a sharp corner in the graph.
4	Limit definition of the derivative	$f'(a) = \lim_{h \rightarrow 0} [f(a+h) - f(a)] / h$; recognizing the pattern identifies f and a .	Evaluating $f(a)$ instead of the slope, or plugging $h = 0$ into the numerator only.	An expression with $(a+h)$ and a division by h .
5	Power and sum rules	d/dx of cx^n is cnx^{n-1} , derivatives of sums are sums of derivatives, and the derivative of a constant is 0.	Keeping the constant term or evaluating the original function instead of the derivative.	A polynomial and "rate of change at $x = \dots$ "
6	Derivatives of exponential functions	d/dx of $Ae^{(kx)}$ is $kAe^{(kx)}$, and d/dx of a^x is $a^x \ln a$.	Forgetting the factor k , or reporting the function value as the rate.	An exponential growth model and "how fast ... at time t ."
7	Equation of the tangent line	The tangent line at $x = a$ is $y = f(a) + f'(a)(x - a)$.	Using the derivative formula itself as the line, or dropping the point.	"Tangent line at $x = a$."
8	Limits at infinity of rational functions	Compare degrees: equal degrees give the ratio of leading coefficients; a smaller numerator degree gives 0; a larger one gives no finite limit.	Using the constant terms, or giving 0 whenever x grows.	"As $x \rightarrow \infty$ " with a polynomial over a polynomial.

Course scope (public description): Calculus I is the study of rates of change in relation to the slope of a curve and covers the knowledge and skills necessary to use differential calculus of one variable and appropriate technology to solve basic problems. Topics include graphing functions and finding their domains and ranges; limits, continuity, differentiability, visual, analytical, and conceptual approaches to the definition of the derivative; the power, chain, and sum rules applied to polynomial and exponential functions, position and velocity; and L'Hopital's Rule. Candidates should have completed a course in Pre-Calculus before engaging in this course.